

Arc synthesis: constructive recovery in dimension one

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

Abstract

Eight tides formalised the constructive side of the germbij note: not merely that Laplace expansion data *determines* the loss (the identifiability arc), but how the data *yields* it. For even-monomial potentials in dimension one the thread certifies four distinct recovery mechanisms — rigidity of power laws, strict monotonicity, the expansion pairing, and uniqueness of limits — and builds the complete expansion ladder behind them: every coefficient of the quartic-perturbed Gaussian partition function, then of the general even-monomial family, explicit with elementary remainders at every order. The final tide crosses to the note’s regular case, reading the seabed’s anharmonic susceptibility limits backwards to certify Theorem 3.1’s triangular induction at its first two rungs. This document synthesises the arc for a reader who will not read eight per-tide retrospectives.

1 The mathematics

The germbij note’s Section 7.4 claims that in dimension one the expansion data of $\int e^{-tL}$ recovers an even-monomial minimum constructively: the leading exponent gives the degree $2k$, the leading coefficient the scale, and the corrections at the graded orders $t^{-(j(m-k)/k+1/(2k))}$ give the subleading coefficients one by one. Its Theorem 3.1 makes the regular-case analogue precise: at a nondegenerate minimum the susceptibilities pair triangularly with the Taylor coefficients. The thread certifies these claims in four mechanisms, ordered by the strength of the data they consume:

Rigidity (exact or asymptotic power-law data): two power functions agreeing on a ray, or asymptotically equivalent at infinity, share exponent and coefficient; hence partition-function data pins the degree k and scale a of a pure even-monomial potential (**Laplace.OneD.Recovery**).

Monotonicity (one temperature): the partition function strictly decreases under pointwise enlargement of the potential strict on positive measure, so for a nonnegative perturbation the coefficient is determined by Z at a *single* t (**Laplace.OneD.RecoveryMonotone**).

The expansion pairing (the note’s mechanism): for $x^2/2 + bx^4$, the $k = 0, 2$ Gaussian moments cancel the main and correction terms exactly, so $Z_b(t) = \sqrt{2\pi}(t^{-1/2} - 3bt^{-3/2}) + O(t^{-5/2})$ with an explicit constant, and b is read off the $t^{-3/2}$ correction by an archimedean argument with no limits (**Laplace.OneD.RecoveryExpansion**).

Uniqueness of limits (susceptibility data): the seabed’s Stage-2 anharmonic limits $(\partial_h \langle x \rangle|_0 \rightarrow -1/\lambda, t \partial_h \text{Cov}|_0 \rightarrow \alpha/\lambda^3)$ read backwards pin λ then α : Theorem 3.1’s induction at orders two and three (**Laplace.OneD.RecoveryAnharmonic**).

Behind the pairing sits the ladder: the exponential Taylor remainder $|e^{-s} - \sum_{j < n} (-s)^j/j!| \leq s^n/n!$ for all $s \geq 0$ (an FTC induction, no other calculus), the all-order quartic expansion with coefficients $\sqrt{2\pi}(-b)^j(4j-1)!!/j!$ at $t^{-(j+1/2)}$, the coefficient-limit packaging (each coefficient is

the limit of the rescaled truncation error), and the potential-generic version for $x^{2k}/(2k)! + b x^{2m}$ with Gamma-form coefficients at exactly the note’s graded orders.¹

2 What made it work

The seabed carried the analysis. Every closed form the thread consumed — the scale- t Gaussian moments, the Gamma-form monomial moments, the monomial integrability family, the Stage-2 anharmonic limits — was built in the May tides for the primer programme. The thread’s own analytic content is one FTC induction and one elementary exponential estimate; everything else is reading existing results in the inverse direction. The productive act in the later tides was the seabed survey, not the proof.

Exact cancellation over approximate bookkeeping. At every order of every ladder, the moment closed forms cancel the expansion terms exactly, so the truncation error *is* the integrated remainder. No estimate is ever made of a quantity that has a closed form.

Elementary bounds over sharp ones. The s^2 remainder constant (double the sharp $s^2/2$) removed all calculus from the first-order tide; the alternating-series route and the Taylor API were both rejected in deliberation for the FTC induction, which uses only stable ingredients. Numerical checks confirmed each constant with the expected slack before any Lean was written — and one check caught invented values that had been written before the script ran, the arc’s one process failure, now documented in the offending tide’s log.

Deliberation amortises. Eight tides used five consult rounds: two candidates were executed on inherited deliberations (the deferred coefficient-limit packaging; the mirror-shaped generic ladder), and the final tide was a first-attempt-clean build because the uniqueness-of-limits pattern had stabilised three tides earlier.

3 Recurring friction

All build iterations in the arc were elaboration-level; the mathematics never moved after deliberation. The catalogue grew three entries, all of one family: combinator-built hypotheses (`Pi.add` integrals, `Pi.div` ratios, `Pi-form` derivative sums) and mixed-cast spellings ($\uparrow(n+1)$ versus $\uparrow n + 1$) fail to match goals syntactically, and the uniform repair is an explicitly-typed or `funext` bridge to the single-lambda, single-cast normal form. `positivity` needs $(x^2)^m$ rather than x^{2m} for symbolic even powers; `rpow` exponent arithmetic wants explicit `show ... by ring` chains, never bare `simp`. The retrospective compile gate (zero errors, zero overfull boxes, checked *before* commit) caught defects in four of the eight tides.

4 What remains

The thread closed its recorded follow-ups. The natural continuations are commissions rather than continuations: the γ rung of Theorem 3.1 needs second-order anharmonic moment expansions (a Stage-2 extension); the full triangular induction needs them at every order; the multivariate Section 7.4 (quasi-homogeneous weight recovery) needs a multivariate moment layer; and the note’s Proposition 4.1 (the flat counterexample) needs expansion theory for smooth non-analytic perturbations. Each is a programme; none is a tide.

¹`Laplace.OneD.RecoveryAllOrder, RecoveryCoeffLimit, RecoveryGeneric.`